^{63}Cu NQR Measurement of Stripe Order Parameter in $La_{2-x}Sr_xCuO_4$

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We demonstrate that one can measure the charge-stripe order parameter in the hole-doped CuO_2 planes of $La_{1.875}Ba_{0.125}CuO_4$, $La_{1.48}Nd_{0.4}Sr_{0.12}CuO_4$, and $La_{1.68}Eu_{0.2}Sr_{0.12}CuO_4$ utilizing the wipeout effects of ^{63}Cu nuclear quadrupole resonance. Application of the same approach to $La_{2-x}Sr_xCuO_4$ reveals the presence of similar stripe order for the entire underdoped superconducting regime $\frac{1}{16} \lesssim x \lesssim \frac{1}{8}$. [S0031-9007(99)09198-X]

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The mechanism of high-T_c superconductivity has been a major controversy throughout the past decade [1–3]. The complexity of the phase diagram for temperature T and hole concentration x makes it difficult to identify the key leading toward the superconducting mechanism. In 1995, Tranquada *et al.* [4–6] demonstrated that $La_{1.6-x}Nd_{0.4}Sr_xCuO_4$ ($x \sim \frac{1}{8}$) exhibits charge-stripe order at $T_{\text{charge}} = 65$ K, followed by a spin-stripe order at somewhat lower temperature, $T_{\text{spin}} = 50$ K. The discovery of the stripe phase [7] has added a new feature to the already complex phase diagram. Initially, some researchers speculated that the stripe order was merely a by-product of the LTO-LTT (low temperature orthorhombic–low temperature tetragonal) structural phase transition and was extrinsic to the fundamental physics of high-T_c superconductivity. However, more recently, spin-stripe order was observed in $La_{2-x}Sr_xCuO_4$ ($x = 0.12$ [8,9] and $x = 0.05$ [10]) and $La_2CuO_{4+\delta}$ [11], two materials without a LTO-LTT structural phase transition. This suggests that the low energy (i.e., *dynamic*) incommensurate spin excitations previously observed for $La_{2-x}Sr_xCuO_4$ over the entire superconducting regime $0.05 \lesssim x \lesssim 0.25$ [12–14] may correspond to the dynamic formation of stripes. Furthermore, recent inelastic neutron scattering experiments have established similar incommensurability of spin excitations also in underdoped $YBa_2Cu_3O_{6.6}$ [15]. These developments raise new hope that understanding the mechanism of stripe instability and its dependence on T and x is the key to establishing a unified picture of the hole-doped CuO_2 planes [1–3].

Unfortunately, stripe order has proved elusive, escaping experimental detection by various techniques. For example, ^{63}Cu nuclear spin-lattice relaxation rate $^{63}1/T_1$ [16,17] measured for another striped material with the LTT structure, $La_{1.88}Ba_{0.12}CuO_4$ [18], exhibits no critical divergence at the stripe transition. The origin of the elusiveness is in the *glassy* nature of the stripe transition [6,18,19]. In other words, the critical slowing down of spin-stripe fluctuations below T_{charge} is more gradual than ordinary magnetic phase transitions involving only the spin degrees of freedom. As a consequence, the apparent critical temperature T_{spin} of the spin-stripe order is lower for experimental probes with slower frequency scales, i.e., $T_{\text{spin}} = 50$ K for elastic neutron scattering ($\sim 10^{11}$ Hz) [4–6], $T_{\text{spin}} = 30$ K for muon spin rotation (μSR, $\sim 10^7$ Hz) [18], and $T_{\text{spin}} = \sim 10$–$30$ K for NQR (nuclear quadrupole resonance, $\sim 10^6$ Hz) [17,20,21].

In this Letter, we propose a new experimental approach to investigate stripes that takes advantage of extreme sensitivity of ^{63}Cu NQR to local charge distribution. We will demonstrate that one can accurately measure the charge-stripe order parameter in $La_{1.875}Ba_{0.125}CuO_4$ and $(La,Nd,Eu)_{1.88}Sr_{0.12}CuO_4$ based on the *wipeout effects* [22] of ^{63}Cu NQR. The basic idea is quite simple. When charge-stripe order sets in at T_{charge}, *rivers* of hole-rich CuO chains separate three-leg CuO ladders without holes, resulting in spatial modulation of hole concentration between 0.5 and 0 holes per Cu atom [4]. This would produce as much as ~ 8 MHz [23] of instantaneous spatial variation of the ^{63}Cu NQR frequency $^{63}\nu_Q$, which is proportional to the electric field gradient [24] at ^{63}Cu nuclei. This means that $^{63}\nu_Q$ is no longer well defined below T_{charge} inside the slowly fluctuating striped domains. Moreover, charge order turns on low frequency spin fluctuations [6], and consequently the ^{63}Cu nuclear spin-lattice and spin-spin relaxation rates $^{63}1/T_1$ and $^{63}1/T_2$, respectively, diverge in the striped domains. All of these effects should wipe out ^{63}Cu NQR signals within stripe ordered domains even before the growth of *static* magnetic hyperfine fields changes the NQR frequency completely below 10 K [17,20]. Therefore, we expect that the wipeout fraction F of ^{63}Cu NQR signals (i.e., the fraction of the lost signal intensity) is a good measure of stripe order. In fact, from comparison with elastic neutron scattering data in $La_{1.48}Nd_{0.4}Sr_{0.12}CuO_4$ we demonstrate that the wipeout fraction F represents nothing but the order parameter for charge stripe. Applying the NQR approach to $La_{2-x}Sr_xCuO_4$, we demonstrate that a similar stripe instability exists over the entire underdoped superconducting regime $\frac{1}{16} \lesssim x \lesssim \frac{1}{8}$, but disappears right above the critical hole concentration $x_c = \frac{1}{8}$.

We grew all of the ceramic samples by standard solid state reactions [17]. The $La_{1.875}Ba_{0.125}CuO_4$ sample was synthesized with 99% ^{63}Cu isotope enriched CuO. We

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conducted the ^{63,65}Cu NQR line shape measurements using the 90°–180° pulse sequence with a fixed separation time t_{90-180}, typically $t_{90-180} = \sim 10-12$ μsec. The ^{63}Cu NQR intensity was estimated by integrating the Gaussian fit of the NQR line shape, with the overall magnitude corrected for the Boltzmann factor and for the spin echo decay time T_2 measured at the NQR peak. The latter process is essential for accurate measurements of the wipeout fraction F because the spin echo decay changes dramatically at T_{charge}. The ^{63}Cu NQR line shape is composed of the so-called A line and B line with the relative intensity of the latter proportional to x, in good agreement with earlier reports [23]. Since high precision measurements conducted for the ^{63}Cu isotope enriched $La_{1.875}Ba_{0.125}CuO_4$ showed identical temperature dependence of the NQR intensity for the A line and the B line between 10 and 300 K, we will focus our attention on the behavior of the more intense A line. The ^{63}Cu NQR intensity decreases dramatically below T_{charge} (i.e., the wipeout fraction F increases from zero to a finite value), but the linewidth of *the observable segments* of the sample (HWHM = $\sim 0.8-1.8$ MHz) showed only a mild increase (up to $\sim 35\%$) with decreasing temperature with no dramatic change at T_{charge}.

In Fig. 1(a), we present the temperature dependence of the wipeout fraction F of the A line in $La_{1.875}Ba_{0.125}CuO_4$, $La_{1.48}Nd_{0.4}Sr_{0.12}CuO_4$, and $La_{1.68}Eu_{0.2}Sr_{0.12}CuO_4$. Within our experimental uncertainties, F is zero between 300 and 65 K (i.e., we detect NQR signals from the entire sample). However, both $^{63}\nu_Q$ and F begin to increase abruptly below $T_{\text{charge}} = 65$ K. $^{63}\nu_Q$ increases ~ 1 MHz between 65 and 10 K showing a similar temperature dependence as F [17], and almost the entire NQR signal disappears by 10 K. We recall that below 10 K, the entire ^{63}Cu resonance intensity reappears as Zeeman perturbed NQR (NQR broadened by static magnetic hyperfine field) between 20 and 80 MHz [17,20]. Coincidence of $T_{\text{charge}} = 65$ K for all three materials with different LTO-LTT transition temperatures [4,19] (~ 65 K for $La_{1.875}Ba_{0.125}CuO_4$ and $La_{1.48}Nd_{0.4}Sr_{0.12}CuO_4$, and ~ 130 K for $La_{1.68}Eu_{0.2}Sr_{0.12}CuO_4$) indicates that the LTO-LTT structural transition is not the primary origin of the observed anomaly at T_{charge}. In fact, $La_{2-x}Sr_xCuO_4$ exhibits qualitatively the same temperature dependence of F without the LTO-LTT structural phase transition, as presented in Fig. 1(b). Accordingly, we must attribute the observed wipeout to electronic effects, as discussed above in the third paragraph. In Fig. 1(a), we compare the wipeout fraction F with the charge- and spin-stripe order parameter determined by elastic neutron scattering experiments [5]. (We note that by definition the elastic neutron scattering intensity represents the square of the order parameter.) *The identical temperature dependence observed for NQR and neutron data for charge ordering allows us to identify the wipeout fraction F as the charge-stripe order parameter.* We note that none

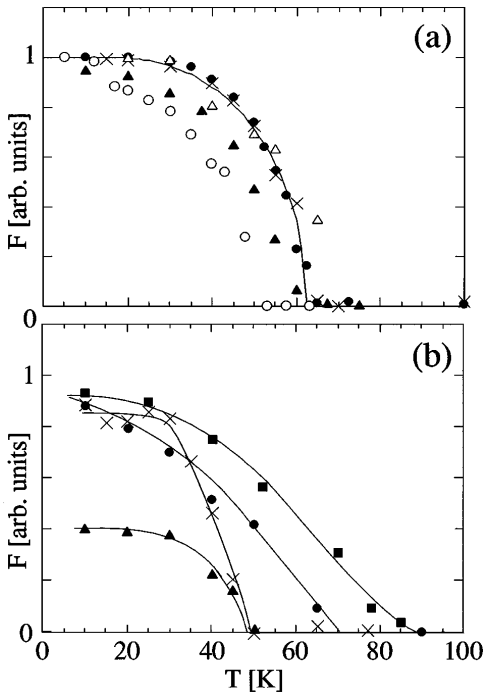


FIG. 1. (a) The wipeout fraction F of ^{63}Cu NQR in $La_{1.875}Ba_{0.125}CuO_4$ (▲), $La_{1.48}Nd_{0.4}Sr_{0.12}CuO_4$ (●), and $La_{1.68}Eu_{0.2}Sr_{0.12}CuO_4$ (×). The solid curve represents the fit to the BCS weak coupling theory with $\Delta = 115$ K. The spin-stripe (○) and charge-stripe (△) order parameter in $La_{1.48}Nd_{0.4}Sr_{0.12}CuO_4$ [3] are also plotted. (b) The wipeout fraction F in $La_{2-x}Sr_xCuO_4$ for $x = 0.125$ (▲), 0.115 (×), 0.09 (●), and 0.07 (■). The curves are guides to the eyes.

of the samples with wipeout effects exhibit divergence of ^{139}La nuclear spin-lattice relaxation rate $^{139}1/T_1$ at T_{charge} [17,25]. Instead, $^{139}1/T_1$ exhibits gradual upturn below T_{charge}, showing a hump below 30 K depending on the hole concentration [17,25]. This is evidence that Cu spin fluctuations do not slow down to NQR frequencies immediately at T_{charge}, in agreement with μSR [18], ESR [19], and earlier NQR line shape measurements [20,21]. We also found that the temperature dependence of the charge-stripe order parameter F fits reasonably well to the weak coupling BCS theory with a gap $\Delta = 1.75 k_B T_{\text{charge}} = 115 \pm 20$ K as shown in Fig. 1(a). This is similar to the case of conventional charge-density wave (CDW) systems [26], and suggests a collective nature of charge-stripe order.

In Fig. 2(a), we present representative results of spin echo decay observed for $La_{1.875}Ba_{0.125}CuO_4$. The results for $La_{1.48}Nd_{0.4}Sr_{0.12}CuO_4$ and $La_{1.68}Eu_{0.2}Sr_{0.12}CuO_4$ were similar [17]. Spin echo decay exhibits a dramatic crossover at T_{charge} from the convolution of Gaussian-Lorentzian [27] to Lorentzian with extremely short $T_2 \sim 15$ μsec ($\ll T_1$ [16,17]), which gradually increases to 10 K. The crossover suggests that above T_{charge} both the Redfield T_1 process (Lorentzian) and the indirect nuclear spin-spin coupling between *like spins* [24] (Gaussian)

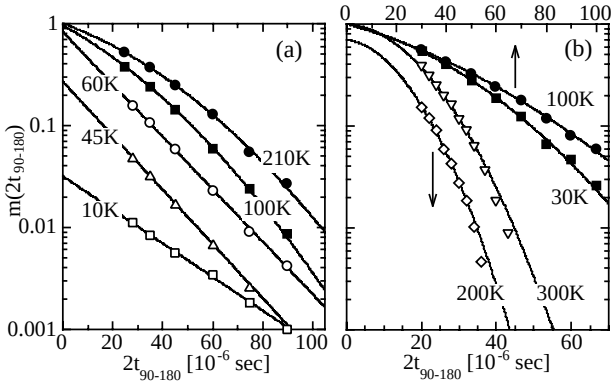


FIG. 2. Spin echo decay normalized by the Boltzmann factor. (a) $La_{1.875}Ba_{0.125}CuO_4$, and (b) $La_{1.85}Sr_{0.15}CuO_4$ (●, ■) and $La_{1.96}Sr_{0.04}CuO_4$ (▽, ◇). Solid curves are the best fits to a convolution of Gaussian-Lorentzian (curved, both the Gaussian and Lorentzian time constants are taken as free parameters) or a pure Lorentzian (straight lines). Notice that the NQR intensity, $m(2t_{90-180} = 0)$ decreases drastically below $T_{\text{charge}} = 65$ K in $La_{1.875}Ba_{0.125}CuO_4$.

contribute to the spin echo decay [27], while below T_{charge} the spatial modulation of charge and $^{63}\nu_Q$ reduces the number of *like spins* and eliminates the indirect coupling effects. The Lorentzian spin echo decay observed below T_{charge} suggests that certain magnetic hyperfine fields H_{hf} at the Cu sites [28] modulate with a time scale τ_c satisfying $\gamma_n H_{hf} \tau_c \ll 1$ [24], where γ_n is the Cu nuclear gyromagnetic ratio. A plausible scenario is that charge stripes fluctuate very slowly even below T_{charge}, resulting in modulation of H_{hf} for the observable segments of the sample. We note that sliding motion of CDW is known to cause similar motional narrowing in conventional CDW systems [29]. The present case is further complicated by the fact that charge order turns on slow spin dynamics [6] and by the fact that we are limited to observing a shrinking domain that has not yet ordered.

We have demonstrated that stripe fluctuations wipe out the ^{63}Cu NQR signal, and the wipeout fraction F is a good measure of the charge-stripe order parameter. Motivated by recent observation of spin-stripe order in $La_2CuO_{4+\delta}$ [11], $La_{1.88}Sr_{0.12}CuO_4$ [8,9], and $La_{1.95}Sr_{0.05}CuO_4$ [10], we have searched for charge-stripe order in $La_{2-x}Sr_xCuO_4$ with $x = 0.15$, 0.135, 0.125, 0.115, 0.09, 0.07, and 0.04. Utilizing the wipeout effects accompanied by crossover to Lorentzian spin echo decay as criteria, we have constructed the phase diagram for charge stripes.

Near the optimum doping level with $x = 0.15$ ($T_c = 38$ K) and 0.135 ($T_c = 35$ K), we observed robust Gaussian behavior as shown in Fig. 2(b), similar to other high-T_c cuprates [27]. A slight change in the form of non-Lorentzian spin echo decay near T_c [30] and the existence of bulk superconductivity make estimation of NQR intensity somewhat tricky, but a very safe estimate of the upper bound of the wipeout fraction F is ~0.3.

The sample with $x = 0.125$ ($T_c = 31$ K) showed pure Lorentzian spin echo decay below $T_{\text{charge}} = 50$ K [not shown, the spin echo decay in the striped phase of $La_{2-x}Sr_xCuO_4$ is semiquantitatively the same as those shown in Fig. 2(a) [17]]. This indicates that stripe fluctuations exist in the entire sample at 50 K. Approximately 40% of the NQR signal disappears by the superconducting transition T_c. The sample with $x = 0.115$ ($T_c = 31$ K) also shows the onset of charge-stripe order at $T_{\text{charge}} = 50$ K, and the striped fraction F exhibits a plateau near T_c, where elastic incommensurate magnetic scattering also sets in [9]. The behavior of samples with smaller hole concentration, $x = 0.09$ ($T_c = 29$ K, $T_{\text{charge}} = 70$ K) and 0.07 ($T_c = 19$ K, $T_{\text{charge}} = 90$ K), is qualitatively similar to the case of $(La,Nd,Eu)_{1.88}Sr_{0.12}CuO_4$ and $La_{1.875}Ba_{0.125}CuO_4$, but the stripe transition is considerably broader. The broad transition may indicate that the stripe is unstable for $x < \frac{1}{8}$. Unfortunately, we cannot determine the real space structure of the stripe [7] based on wipeout effects alone. The Meissner fraction in $x = 0.125$, 0.115, 0.09, and 0.07 [17] is comparable to the observable NQR fraction, $1 - F$, at T_c for each sample. Whether superconductivity coexists with stripe, or microscopic phase separation sets in at T_c, is a matter of debate. However, it is certainly true that a canonical superconducting transition with a large specific heat jump at T_c is realized only above $x = 0.125$ [31].

In contrast to the superconducting samples, the lightly hole-doped nonsuperconducting $La_{1.96}Sr_{0.04}CuO_4$ showed a gradual decrease of the NQR intensity below 300 K. However, the spin echo decay always has a Gaussian component as shown in Fig. 2(b), and there is no evidence for stripe fluctuations. This is consistent with the neutron scattering result that spin response is commensurate below $x = 0.05$ without the signature of spin stripes [10,13,32]. Since dc resistivity for $x = 0.04$ is large and shows a gradual upturn due to weak localization below 300 K [32], we attribute the gradual loss of the NQR signal to commensurate short-range spin order in the vicinity of *randomly* localized holes. In this context, it is worth recalling that even conventional dilute Kondo alloys such as Cu:Fe exhibit similar gradual wipeout ($F \sim \frac{1}{T}$) above the Kondo temperature [33].

In Fig. 3 we summarize the critical temperature T_{charge} and order parameter F as a function of x. Clearly, the ground state of $La_{2-x}Sr_xCuO_4$ exhibits a sharp crossover at the critical hole concentration $x_c = \frac{1}{8}$ from the striped state to the superconducting state with low energy (~meV) stripe fluctuations [12–14]. The charge-ordering temperature T_{charge} is consistently higher than the spin-ordering temperature determined by elastic neutron scattering ($T_{\text{spin}} = 30$ K for $x = 0.12$ [9], and 17 K for $x = 0.05$ [10]), μSR and ^{139}La NQR ($T_{\text{spin}} \sim 15$ K is the largest for $x = 0.115$ [17,18,21,25]). Perhaps remnant fluctuations of charge stripes prevent spins from freezing, resulting in the glassy transition. The fact that

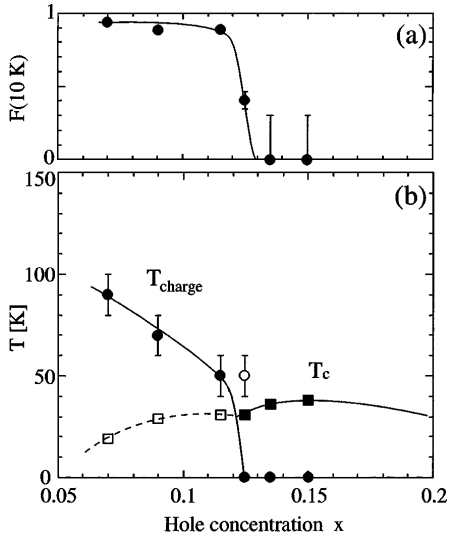


FIG. 3. (a) The charge-stripe order parameter F at 10 K in $La_{2-x}Sr_xCuO_4$. (b) The phase diagram of the charge-stripe order T_{charge} (●, ○) and the superconducting critical temperature T_c (■, □) in $La_{2-x}Sr_xCuO_4$. Filled (open) symbols denote the phase boundary with more (less) than 50% fraction.

T_{charge} is much higher than T_{spin} in $La_{2-x}Sr_xCuO_4$ may indicate that the charge stripes are significantly disordered. The previously undetected phase boundary of charge stripe in Fig. 3(b) explains why insulating (i.e., divergent) behavior of $^{63}1/T_1$ [34] and dc resistivity [35] crosses over to metallic in the vicinity of the optimal superconducting phase $x = 0.15$. These observations as well as recent neutron data [4–6,8–14] are compatible with, but not necessarily a proof of, a viewpoint [3,36] that stripe fluctuations near $x_c = \frac{1}{8}$ have a connection with the mechanism of high-T_c superconductivity. Even if the stripe instability is actually disfavorable for superconducting pairing, the proximity between the quasistatic stripe phase and the canonical superconducting phase in Fig. 3(b) indicates that they are energetically very close. In any case, the implications of all earlier experiments reported for underdoped, superconducting $La_{2-x}Sr_xCuO_4$ ($\frac{1}{16} \lesssim x \lesssim \frac{1}{8}$) during the past decade must be reassessed in light of our finding that there exist slowly fluctuating, quasistatic charge stripes.

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